Shashwat Raj

Applied AI Engineer
Mumbai, India

shashwatrajiitd@gmail.com
+91 9682638110
linkedin.com/in/shashwatrajiitd/
www.shashwatraj.in

Year	Degree / Board	Institute
2025	Bachelor of Technology (B.Tech.) in Mathematics & Computing	Indian Institute of Technology (IIT), Delhi
2021	CBSE (Computer Science, Mathematics, Physics)	Stephen International School, Jammu

WORK EXPERIENCE

- **Applied AI Engineer** | *Purplle (Manash Lifestyle Pvt. Ltd.)* *(April 2025 – Present)*
  - Architected an autonomous **Multi-Agent Creative Generation System** for UGC, **Meta & Google** paid channels.
  - Built a **microservice pipeline (AI Agents Orchestration)** producing **1,000+ pixel-perfect variants** from single PSD.
  - **Inference Optimization:** Prompt compression, caching & batch API; reduced per-creative GenAI spend by **60%**.
  - **Creative Bank:** Centralised Master Content Bank (MCB) for Marketing, Product, Central Design, & Brand Teams.
  - **Expanding:** Phase-I targets Agentic-personalisation across user journeys with built-in business objectives & KPIs.
  - **Tech stack:** Gemini, Diffusion, LangChain, ADK, FastAPI, Redis, Kafka, MySQL, Qdrant, Skia, Docker, GCP.
  - **Business Impact:** *Generating **3,000+ ideas** and **1,500+ creatives/month** across **7 brands**; turnaround **10–15**$\times$ faster, cost reduced by $\sim$**95%**; replacing Rocketium (SaaS Cost), saving **INR 60L+/year** in subscription costs.*

INTERNSHIPS

- **Software Developer** | *Samsung R&D Institute, Bangalore* *(May 2024 – July 2024)*
  - Built a **Revenue forecasting & Anomaly Detection event-based Pipeline** for Ad-PF team with **anomaly detection**.
  - Improved system observability & **increased forecast accuracy by 15%** across high-volume revenue data streams.
  - **Deployed Transformer-based forecasting models** (TFT) with **LSTM/GRU ensembles** for multivariate prediction.
  - **Tech stack:** Python, R, MySQL, Forecasting Models (Linear & Transformer), AWS, Data processing, Apache Kafka.

TECHNICAL SKILLS

- **Programming & Systems**: Python, SQL, C/C++, GCP, Docker, REST API, Distributed Systems, System Design.
- **GenAI & AI Models**: LLM, OpenAI/Gemini, RAG, Agentic Systems, Vector DB, Prompt Engineering, Computer Vision.
- **Frameworks & Tooling**: FastAPI, LangChain, LangGraph, Google ADK, Redis, Qdrant, Kafka, Celery, Pydantic, vLLM.
- **MLOps & Infrastructure**: Kubernetes, Docker, CI/CD, Model Serving, Observability, Cost Optimization, Scalability.

PROJECTS

- **Autonomous Social Media AI Agent** | *Personal Applied AI Systems Project* *(2026)*
  - Designed a self-hosted **multi-agent pipeline** for autonomous content discovery across X, Reddit, Instagram PFs.
  - Built trend ingestion via **embedding similarity**; integrated with **Helix - Creative Generation Agent Orchestration.**
  - **Tech stack:** Vector DB, CLIP, LLM, Playwright, Docker, SQLite/JSON, Pydantic, PostgreSQL, Agent Orchestration
  - **Impact:** Generating **5+ creatives/day**, running pages autonomously with **1000+ engagements per day per content.**
- **Netflix-Style Portfolio Platform** | *shashwatraj.in* | *Full-Stack AI Systems Project* *(2025)*
  - Built a **profile-driven monorepo** (Next.js + FastAPI) with Netflix-style UI, drag carousels, and SSE streaming.
  - Engineered **RAG pipeline** using ChromaDB vector store and Gemini embeddings for profile-aware grounded answers.
  - Designed **sandboxed Python code execution** backend streaming stdout/stderr in real-time via SSE to Monaco IDE.
  - **Tech stack:** Next.js, FastAPI, ChromaDB, Gemini, Docker, Cloud Run, Turborepo, TypeScript, Python, Nginx.
- **OS Internals: Scheduler & UNIX Shell** | *Systems Programming Project* *(2024)*
  - Implemented **FCFS, Round Robin, and MLFQ** (3-level priority queue) as offline CPU scheduling algorithms in C.
  - Built **online schedulers** (SJF, SRTF, adaptive MLFQ) with historical burst-time tracking and priority adjustment.
  - Developed a **UNIX-like shell** in C with pipe support, exec syscalls, built-in cd/history, and signal handling.
  - **Tech stack:** C, POSIX, fork/exec, IPC Pipes, Signal Handling, Process Scheduling, Linux Systems Programming.
- **Memory Management Unit (MMU) Simulator** | *Systems Programming Project* *(2023)*
  - Built a MMU simulator in C managing **4GB virtual memory** via single-level page tables and TLB emulation.
  - Implemented process lifecycle: creation, forking, memory allocation, deallocation, and graceful termination.
  - Simulated address translation, page faults, and frame eviction using **LRU replacement policy** in C.
  - **Tech stack:** C, Virtual Memory, Page Tables, TLB, LRU Eviction, Process Forking, Linux Systems.

ACHIEVEMENTS

- **JEE Advanced & Mains 2021**: Secured **99.89 percentile** nationwide (top 0.2%) among **1M+** candidates across India.
- **Samsung SWC Test (2025)**: Cleared Advanced Software Competency Test at Samsung R&D, Bangalore.
- **Regional Mathematical Olympiad**: Awarded certificate for mathematical excellence twice by HBCSE, India.
- **NSEP Merit Certificate**: Achieved Merit Certificate for ranking in the **Top 1%** nationwide in Physics Olympiad.
- **KVPY Fellowship 2021**: Awarded prestigious DST, GOI fellowship (Stage 1) for ranking in the **Top 1%** nationally.
- **NTSE Scholar 2019**: Awarded National Talent Scholarship, selected among **1M+** students nationwide.

POSITIONS OF RESPONSIBILITY / LEADERSHIP

- **Coordinator, RDV (Rendezvous) '23, BRCA** – Organized and managed cultural events *(Jun 2023 – May 2024)*
- **Class Representative** to HWC (House Working Committee), Karakoram Hostel, BHM *(Jun 2022 – May 2023)*